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MODEL SELECTION AND VALIDATION REPORT ( pielou_biomass model )
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## OBSERVATIONS:3451

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( pielou_biomass model )

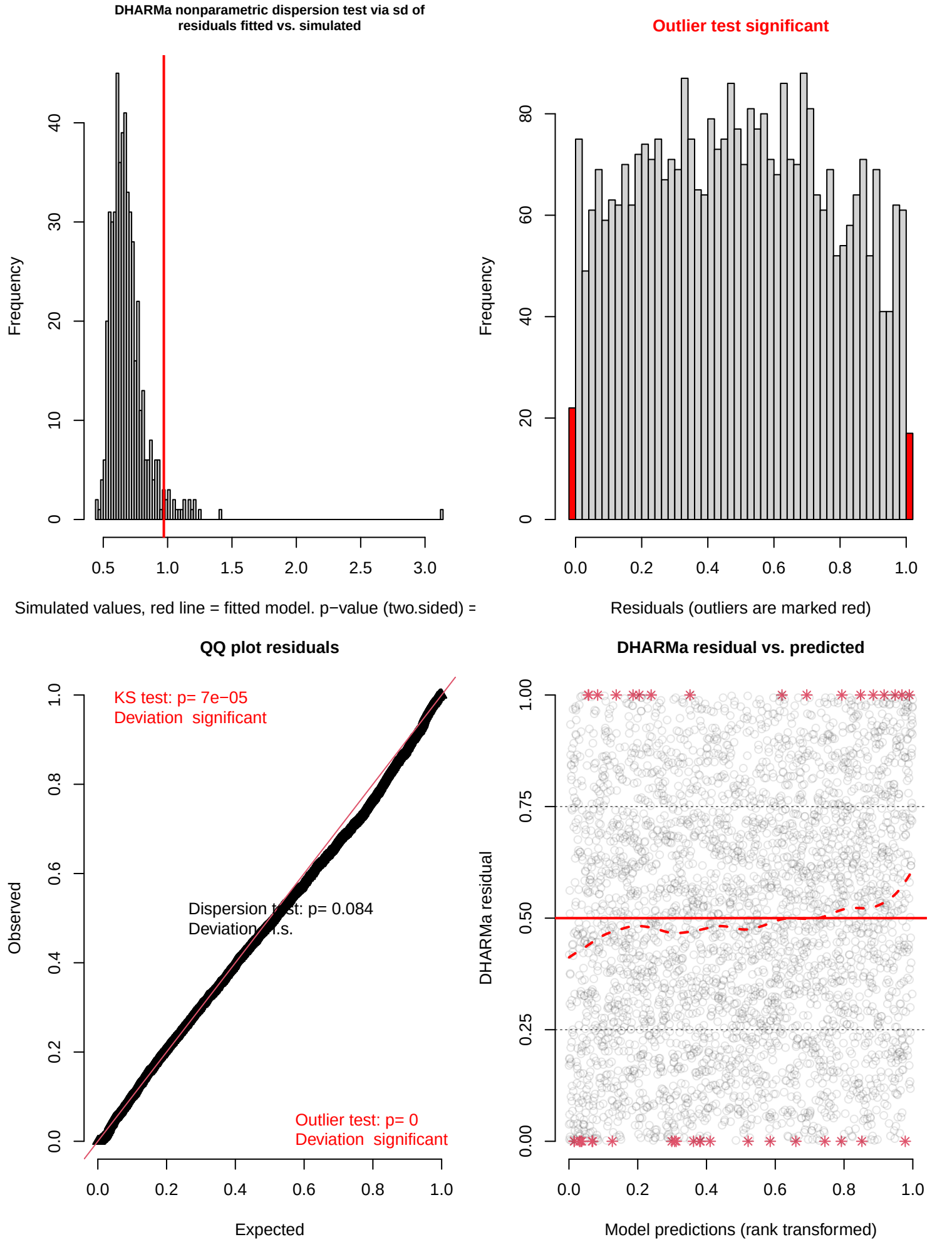
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: lognormal(link = 'log')

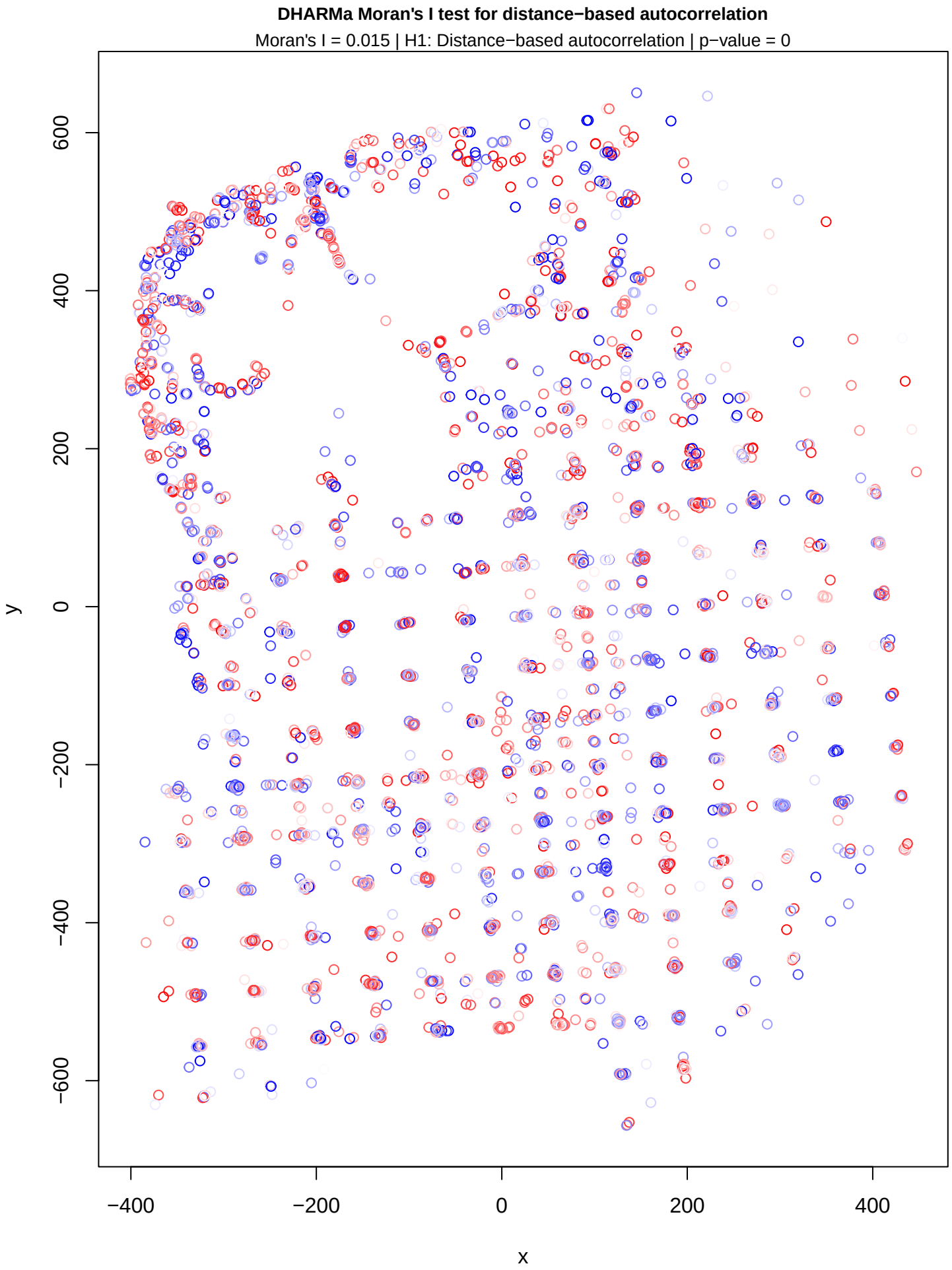
Conditional model:
              coef.est coef.se
(Intercept)       7.01    0.19
kde_fric          0.10    0.02
kde_feve         −0.11    0.02
tric              0.08    0.02
p_teve           −0.42    0.02
sbt_mean          0.34    0.05
sic_mean          0.03    0.05
sbt_sd            0.11    0.03
sst_sd            0.04    0.03
log_chla_mean     0.02    0.02
depth             0.01    0.03

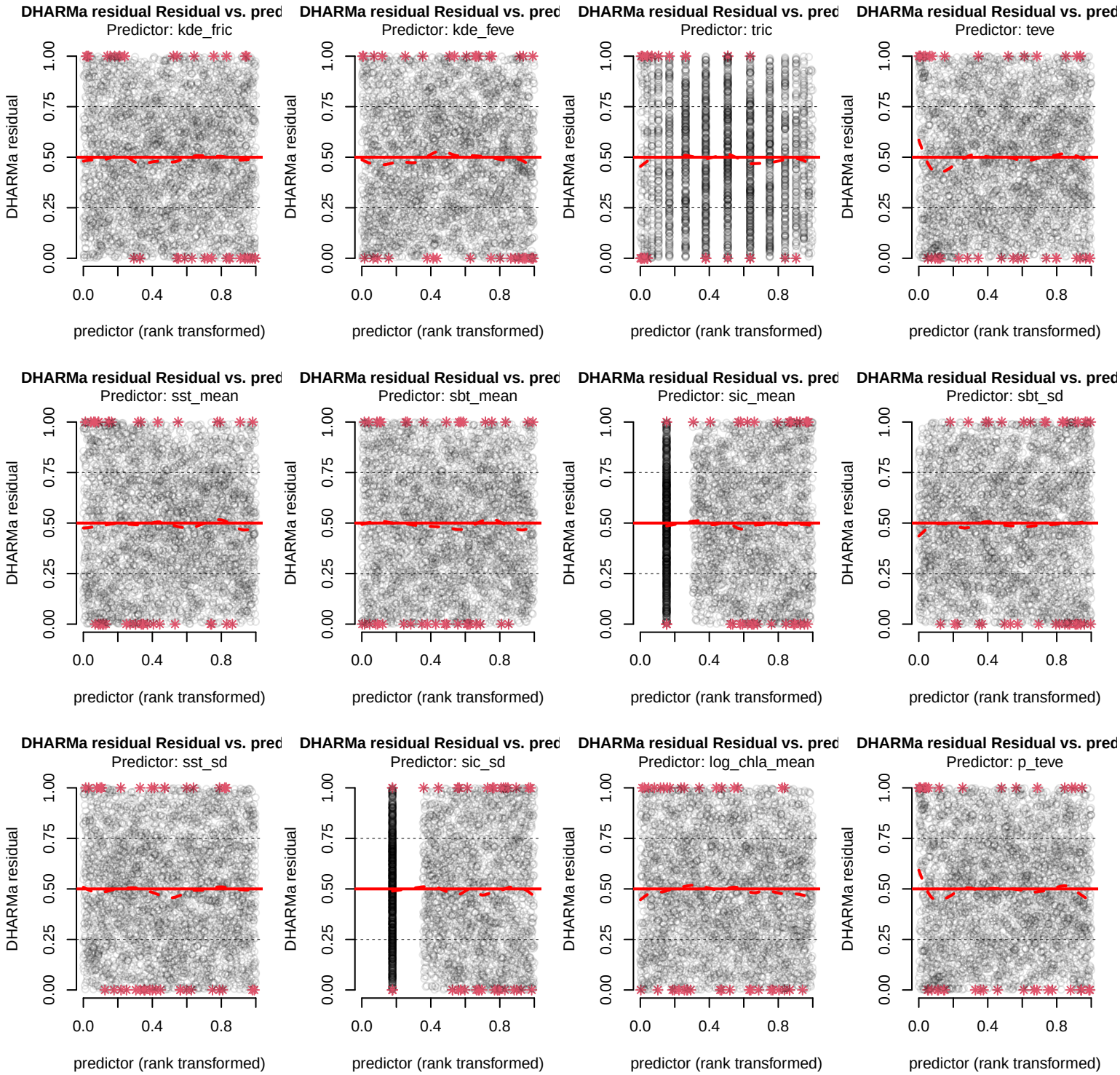
Dispersion parameter: 0.74
Spatiotemporal AR1 correlation (rho): 0.87
Matérn range: 299.22
Spatiotemporal marginal AR1 SD: 0.85
REML criterion at convergence: 27536.615

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                   df      AIC   delta_AIC
spatiotemporal_random_fields_ar1_with_anisotropy 17 55103.02   0.2113781
spatiotemporal_random_fields_ar1                 15 55103.23 300.7756374
spatiotemporal_random_fields_iid_with_anisotropy 16 55404.01   5.2523280
spatiotemporal_random_fields_iid                 14 55409.26 330.2631877
spatial_random_fields                            14 55739.52   3.7658459
spatial_random_fields_with_anisotropy            16 55743.29 677.4163477
error_only                                       12 56420.70          NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
Simulated values, red line = fitted model. p−value (two.sided) =
Outlier test significant
Frequency
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 7e−05
Deviation significant
Dispersion test: p= 0.084
Deviation n.s.
Outlier test: p= 0
Deviation significant
Observed
Expected
DHARMa residual vs. predicted
DHARMa residual
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.015 | H1: Distance−based autocorrelation | p−value = 0
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. pred
Predictor: kde_fric
Predictor: kde_feve
Predictor: tric
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
Predictor: p_teve
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8